

Akashdeep Guruswamy

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EDUCATION

New Jersey Institute of Technology

January 2020 - December 2021

Master of Science in Computer Science

PES University

August 2014 - May 2018

Bachelor of Technology in Mechanical Engineering

EXPERIENCE

Software Engineer

May 2022 - October 2024

SecureGen Inc, New Brunswick, NJ

Led end-to-end development of secure, cloud-based research data applications for pharmaceutical organizations.

- **Boosted platform security** by implementing two-factor authentication and real-time email/SMS alerts in React, reducing unauthorized access incidents by 40%.
- **Developed a Python package and accompanying UI** for ML model analysis and visualization—resulting in 30% faster model iteration by end-users.
- **Designed model run comparison tools** that increased model-sharing engagement by 60% across the research team.
- **Contributed to regulatory compliance** by co-developing a secure licensing system for version control tools used in clinical environments.
- Ensured system reliability through automated testing using **Selenium**, enhancing test coverage and reducing bugs by 35%.

Software Engineer Co-op

September 2021 - December 2021

Teledyne Lecroy, Chestnut Ridge, NY

Worked in a fast-paced agile team to modernize and streamline internal software infrastructure.

- **Migrated legacy logging systems** to a modern, real-time logging architecture using **Node.js, Docker, and Kubernetes**, reducing transcription delays by 60%.
- **Integrated Five9 API** with internal transcription tools, improving automation and minimizing manual effort in probe reading conversion.
- Participated in daily standups, sprint planning, and code reviews, contributing to a 15% faster sprint cycle completion.

PROJECTS

Mobile Offloading

January 2021 - May 2021

- Engineered a fault-tolerant distributed computing model using Android devices in a master-slave topology.
- Demonstrated 25% performance improvement over centralized systems for large datasets.
- Tools/Tech: Java, Android Studio, Hadoop, Spark

Fairness on Machine Learning Models

September 2021 - December 2021

- Analyzed Lymphography dataset to explore trade-offs between model **accuracy and fairness**.
- Built predictive models and interactive visualizations using **Python, Plotly, ggplot**, achieving **89% accuracy with Random Forest**.
- Tools/Tech: Python, Jupyter Notebook, Plotly

TECHNICAL SKILLS AND INTERESTS

TypeScript | JavaScript | HTML / CSS | Python | Bash | SML | SQL | Java | React | Node.js | Flask | MongoDB | Celery |

React Native | Django | Spring Boot | AWS | MySQL | Git